

Data Science for Social Good

Project Proposal

Team Members:

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Social Problem

When the cost of essential raw materials and energy spikes, businesses face immediate financial strain. As cutting wages in response would be difficult, companies often protect their bottom line by freezing hiring or even firing workers. Managing these laid off workers can pose a challenge for the government, which can be particularly severe after a massive recession. Policymakers need time and information to properly plan how to absorb this influx of workers, however, official government unemployment reports are only published weeks after these job losses have already occurred. This project aims to build an early warning system that predicts unemployment spikes before they are officially reported, so that safety-net resources can be allocated proactively. It will also predict the number of weeks it takes for the commodity price spike to significantly translate into a spike in unemployment claims.

For this, we will be analyzing price fluctuations in physical commodity markets rather than stock market data. This is because the prices of physical commodities like oil, gas, iron ore and copper represent the actual cost of running a business, whereas stock markets are often driven by investor opinion. Thus, we believe commodity price shocks are a more direct and reliable cause of worker layoffs.

SDG

This project addresses SDG 8: Decent Work and Economic Growth, specifically Target 8.5: Full Employment and Decent Work with Equal Pay. It does so by providing policymakers with the information necessary to take measures to absorb the influx of recently laid-off workers into the job market, preferably without wage reduction.

Initial Hypothesis

Sudden price spikes and volatility in core industrial (Copper and Iron Ore) and energy commodities (Crude Oil and Natural Gas) will predict an increase in US Initial Jobless Claims. However, this reaction will not be immediate as it takes time for businesses to absorb these costs, review their budgets, and make the decision to officially let workers go. We expect to find a lag of several weeks between the initial price spike and the resulting unemployment claims.

Dataset

- **Sources:** Our primary data source will be the US government's Federal Reserve Economic Data (FRED) API. We will use it for weekly US Initial Jobless Claims, as well as daily commodity prices for Crude Oil and Natural Gas, which will be aggregated to a weekly frequency. Since FRED only publishes Copper and Iron Ore prices monthly, we will run a hypothesis test comparing FRED's monthly prices against Python's 'yfinance' library's daily prices for these two metals. If the test doesn't show a statistically significant difference, we will use yfinance for Copper and Iron Ore daily prices, aggregated to weekly. If the difference is significant, we will use FRED's monthly prices, converted to a weekly, to preserve data reliability.
- **Curation Effort:** We will merge daily and monthly commodity prices data with weekly government labor reports using the necessary pre-processing aggregation and averaging. We will also create multiple additional variables that store how much prices changed up to 16 weeks ago so the ML model can discover the exact delay between a price shock and the resultant layoff.
- **Scope:** To train our model, we will collect data from 2007 to September 2026. This would give us a dataset comprising roughly 1000 weeks of historical data, including historically significant recessions such as the 2008 financial crash and the 2020 pandemic lay-offs. Our model will initially use variables such as Base Weekly Prices, Volatility Features, and percentage price shocks at up to 16 weeks before the layoff for each commodity. This wide lag window will only be exploratory, and we will keep the strongest lags per commodity as final features to keep variable count manageable relative to dataset size. Using this data, our model will predict the weekly count of unemployment claims and the number of weeks it takes for the commodity price spike to significantly translate into a spike in unemployment claims.
- **Limitations:** Occasionally, commodity prices spike not because of real-world shortages but due to Wall Street traders' speculations. This could occasionally trick our model into predicting layoffs that do not actually occur. During the 2008 financial crash and the 2020 pandemic lay-offs, unemployment spiked due to reasons unrelated to commodity pricing, and we will decide after analyzing the model's result whether accuracy would be improved by keeping or removing data from these time periods.

Future Extension

As we will be collecting data from live sources, extending the dataset is part of our plan. In Week 14, we will collect the past 9 weeks of new data that were generated during the semester. We will test our model by feeding it only the new commodity prices to see whether it predicted the actual unemployment numbers reported by the US government over those 9 weeks (the accuracy of our prediction will be assessed through Trend Matching and Mean Absolute Error).

This newly acquired data will be incorporated into our original dataset and be used to retrain the model to see if the 'rules of the economy' have changed in the past 9 weeks, i.e. do businesses lay off workers faster or slower than they did earlier in response to price shocks?